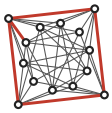
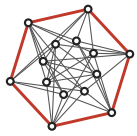
M?b@bFHNEk@~|k}U?
44 edges · $H_1 = 0$
G14a: a = 5-cycle



M?B@thku?~Lylk}[?
45 edges · $H_1 = 0$
G14b: a = 4-cycle + spur



M?r@d`ifbiFfo}~o?
45 edges · $H_1 = 0$
G14c: a = 6-cycle



Green box: the 3 contractible stuck graphs G14a, G14b, G14c (I-contractible, not strong I-contractible; word $a \cdot a \cdot a^{-1}$ on the cycle a). Below: the 48 stuck graphs with $H_1(\Delta; \mathbb{Z}) = \mathbb{Z}/3$, by edge count. Red: edges in $\neq 2$ triangles; for discs, the identification cycle a (in G14b with a spur). * Annulus: a 4-cycle and a disjoint edge in 4 triangles. ** Annulus and disc: two disjoint 4-cycles. All 51 are K_4-free, $\mathbb{Z}_2$-acyclic, with no I_S-contractible vertex or edge; all such graphs under H14 (geng -k -C -d4 -D12 14).

M?B@cXamAu\r{]V{?
43 edges · $H_1 = \mathbb{Z}/3$



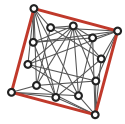
M?B@cXemAuB}ze{]?
43 edges · $H_1 = \mathbb{Z}/3$



M?BDA_wqpmJvj]|w?
43 edges · $H_1 = \mathbb{Z}/3$



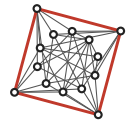
M?`@E_wDf[UnU}z{?
43 edges · $H_1 = \mathbb{Z}/3$



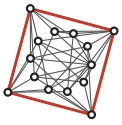
M?`adQWT\LK^du^w?
43 edges · $H_1 = \mathbb{Z}/3$



M?`adQWtVDUr[]^w?
43 edges · $H_1 = \mathbb{Z}/3$



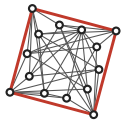
M?bB@_WmbRJZ|M~w?
43 edges · $H_1 = \mathbb{Z}/3$



M?bB@_WsbX^Jlm~w?
43 edges · $H_1 = \mathbb{Z}/3$



M?bB@_WsequuZZ~w?
43 edges · $H_1 = \mathbb{Z}/3$



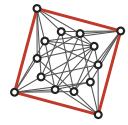
M?bB@_Wu`ZJjlm~w?
43 edges · $H_1 = \mathbb{Z}/3$



M?bB@_WubXNJlm~w?
43 edges · $H_1 = \mathbb{Z}/3$



M?bB@aWYTkZZ^M|w?
43 edges · $H_1 = \mathbb{Z}/3$



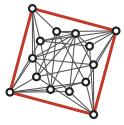
M?bB@aWZSmXZ|wNm?
43 edges · $H_1 = \mathbb{Z}/3$



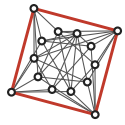
M?bB@aWZTkFf|wuu?
43 edges · $H_1 = \mathbb{Z}/3$



M?bB@aWdbYFv|wuu?
43 edges · $H_1 = \mathbb{Z}/3$



M?bB@aWdbZEv|wuu?
43 edges · $H_1 = \mathbb{Z}/3$



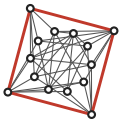
M?bB@aWqplQzZ]|w?
43 edges · $H_1 = \mathbb{Z}/3$



M?bB@aWyTkYZ^M|w?
43 edges · $H_1 = \mathbb{Z}/3$



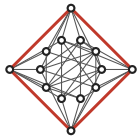
M?bB@aXdbY[{Nmuu?
43 edges · $H_1 = \mathbb{Z}/3$



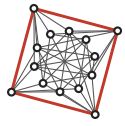
M?bB@aXlbYFfxwuu?
43 edges · $H_1 = \mathbb{Z}/3$



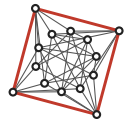
M?bB@aXyTkYZxw^M?
43 edges · $H_1 = \mathbb{Z}/3$



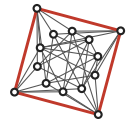
M?otU`SmVEKm]M^o?
43 edges · $H_1 = \mathbb{Z}/3$



M?qa``{rfEEymitY?
43 edges · $H_1 = \mathbb{Z}/3$



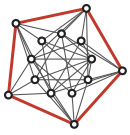
M?qa``{ufEUimitY?
43 edges · $H_1 = \mathbb{Z}/3$



M?r@daiRbrTYd{tb?
43 edges · $H_1 = \mathbb{Z}/3$



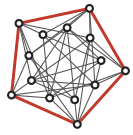
M?`Db_kVcyl]hyvw?
44 edges · $H_1 = \mathbb{Z}/3$



M?`Db_khi{L\nUvw?
44 edges · $H_1 = \mathbb{Z}/3$



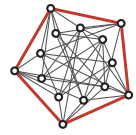
M?`adL[mBkvLY{d{?
44 edges · $H_1 = \mathbb{Z}/3$



M?`cu_whRRFZ~Wtm?
44 edges · $H_1 = \mathbb{Z}/3$



M?`cv@UNDNZMtsY{?
44 edges · $H_1 = \mathbb{Z}/3$



M?b@bCw`usLvvw{]?
44 edges · $H_1 = \mathbb{Z}/3$



M?qa``[NNEYlme~o?
44 edges · $H_1 = \mathbb{Z}/3$



M?qa``[iUEylNm~o?
44 edges · $H_1 = \mathbb{Z}/3$



M?qa``[icjk^Nk~o?
44 edges · $H_1 = \mathbb{Z}/3$



M?qa``[iej[ZNk~o?
44 edges · $H_1 = \mathbb{Z}/3$



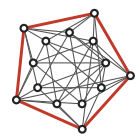
M?qa``kLeE^wt]i}?
44 edges · $H_1 = \mathbb{Z}/3$



M?qa``kMmMIniu~o?
44 edges · $H_1 = \mathbb{Z}/3$



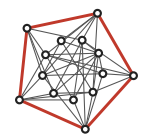
M?qa``kifEfUtZ~o?
44 edges · $H_1 = \mathbb{Z}/3$



M?qa`awkmiNdwu~o?
44 edges · $H_1 = \mathbb{Z}/3$



M?qa`awwRxYj{k~o?
44 edges · $H_1 = \mathbb{Z}/3$



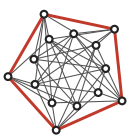
M?qa`g{guLzwmkU]?
44 edges · $H_1 = \mathbb{Z}/3$



M?qa`jSwubncvomd?
44 edges · $H_1 = \mathbb{Z}/3$



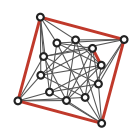
M?qaejaLe[ngmh{[?
44 edges · $H_1 = \mathbb{Z}/3$



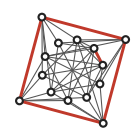
M?`FSzcVBmVFs{J{?
45 edges · $H_1 = \mathbb{Z}/3$



M?qa``{ws}Lxmkt[?
45 edges · $H_1 = \mathbb{Z}/3$ *



M?qa``{wvkVUt[}Q?
45 edges · $H_1 = \mathbb{Z}/3$ *



MCOfczgVB]UeM{p{?
45 edges · $H_1 = \mathbb{Z}/3$



MCp`faiFf{W}][js?
47 edges · $H_1 = \mathbb{Z}/3$ **

